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Design and Implementation of an Op-Amp-Based KHN Audio Filter

Submitted in partial fulfillment of the requirements

of the degree of

Bachelor of Technology

(Electronics and Telecommunication Engineering)

by

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CERTIFICATE

This is to certify that the project report entitled

'Design and Implementation of an Op-Amp-Based KHN Audio Filter'

submitted by

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Dr. Deeplaxmi Niture

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DECLARATION

I/We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

This project presents the design and implementation of an Op-Amp-Based KHN (Kerwin–Huelsman–Newcomb) Audio Filter for audio signal processing and active noise cancellation applications. The KHN state-variable filter topology generates simultaneous Low-Pass, Band-Pass, and High-Pass outputs using operational amplifiers. The filter is designed around a center frequency of approximately 1 kHz and demonstrates second-order filter characteristics suitable for educational and audio engineering applications.

The project includes theoretical analysis, mathematical modelling, circuit calculations, Proteus simulations, Bode plot analysis, step response analysis, and hardware implementation using TL071 op-amps. The results obtained through simulations and Digital Storage Oscilloscope (DSO) measurements confirm the expected frequency response and stability of the KHN filter.

KEYWORDS

KHN Filter, Active Filter, Operational Amplifier, Audio Signal Processing, State Variable Filter, Low Pass Filter, Band Pass Filter, High Pass Filter, Bode Plot, Active Noise Cancellation

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ABBREVIATIONS AND NOMENCLATURE

Abbreviation	Meaning
KHN	Kerwin–Huelsman–Newcomb
LPF	Low Pass Filter
BPF	Band Pass Filter
HPF	High Pass Filter
DSO	Digital Storage Oscilloscope
ANC	Active Noise Cancellation
Op-Amp	Operational Amplifier
f_0	Center Frequency
Q	Quality Factor

1. INTRODUCTION

This project focuses on the design and implementation of a KHN active filter using operational amplifiers. The KHN filter is widely used in audio applications because it simultaneously produces Low-Pass, Band-Pass, and High-Pass outputs. Active filters are preferred over passive filters due to better stability, gain control, and frequency response characteristics. The filter designed in this project operates around a center frequency of 1 kHz and demonstrates second-order behavior suitable for audio processing and educational experiments.

2. AIM AND OBJECTIVES

Primary Aim:

To design and implement an Op-Amp-Based KHN Audio Filter.

Objectives:

- Design a second-order KHN state-variable filter.
- Generate LPF, BPF, and HPF outputs.
- Verify filter response through Proteus simulations.
- Analyze Bode plots and transient response.
- Implement the circuit on hardware and verify outputs using DSO.

3. SPECIFICATIONS AND REQUIREMENTS

Filter Type: KHN State Variable Filter

Center Frequency: 1 kHz

Outputs: LPF, BPF, HPF

Power Supply: $\pm 12\text{V}$

Active Device: TL071 Operational Amplifier

Frequency Range: 20 Hz – 20 kHz

4. BLOCK DIAGRAM

The op-amp-based KHN filter consists of three main blocks:

1. Summing Amplifier (Op-Amp 1) – generates the high-pass output by combining input and feedback signals.
2. First Integrator (Op-Amp 2) – integrates the high-pass signal to generate the band-pass output.
3. Second Integrator (Op-Amp 3) – integrates the band-pass signal to generate the low-pass output.

Feedback from the BP and LP outputs is fed back to the summing amplifier to establish the overall second-order behavior.

Op-Amp-Based KHN Audio Filter Block Diagram

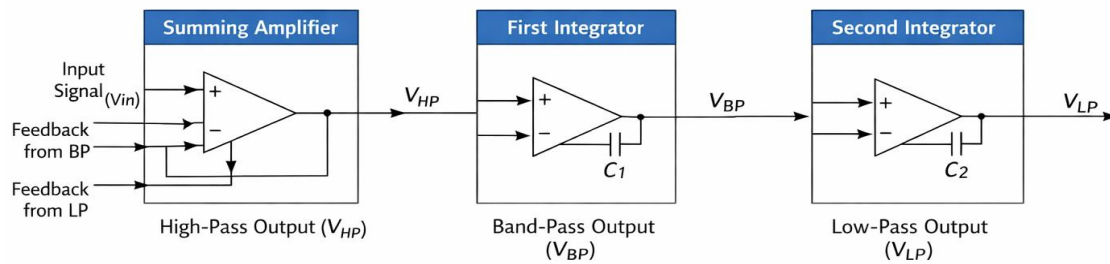


Fig 4.1 – Block Diagram of KHN Filter

5. WORKING PRINCIPLE AND THEORY

The KHN filter is based on state-variable filter topology. The first op-amp acts as a summing amplifier and combines the input signal with feedback signals. The second and third op-amps act as integrators. By controlling the resistor and capacitor values, the center frequency and quality factor can be adjusted independently. The filter exhibits second-order characteristics and stable frequency response.

6. CIRCUIT ARCHITECTURE

The op-amp-based KHN filter uses:

- One summing amplifier
- Two integrators
- Two capacitors (C1 and C2)
- Five resistors controlling gain and feedback

Each integrator uses a resistor-capacitor (RC) network in the feedback loop of an op-amp, ensuring stable and predictable integration.

The summing amplifier combines the input signal with feedback from the BP and LP outputs to create the correct high-pass behavior.

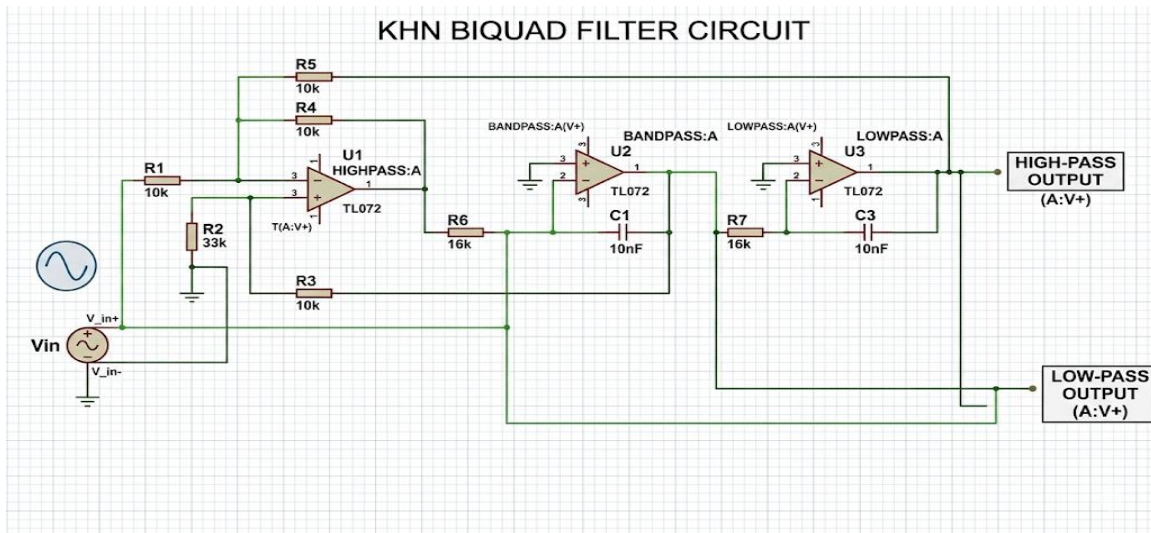


Fig 6.1 – Complete KHN Circuit

7. CIRCUIT DESIGN AND CALCULATIONS

The standard denominator equation of the second-order filter is:

$$D(s) = s^2 + (\omega_0/Q)s + \omega_0^2$$

Using:

$$R = 16 \text{ k}\Omega$$

$$C = 10 \text{ nF}$$

$$f_0 = 1/(2\pi RC)$$

$$\approx 995 \text{ Hz} \approx 1 \text{ kHz}$$

Equal resistor values were selected in the summing amplifier stage to maintain unity gain and balanced response. The quality factor was adjusted using resistor ratios to obtain Butterworth characteristics.

Detailed Design Calculations:

The KHN filter uses two cascaded integrator stages. The center frequency depends upon the RC time constant.

Given:

$$R = 16 \text{ k}\Omega$$

$$C = 10 \text{ nF}$$

$$f_0 = 1 / (2\pi RC)$$

$$f_0 = 1 / (2\pi \times 16 \times 10^3 \times 10 \times 10^{-9})$$

$$f_0 = 995 \text{ Hz}$$

The quality factor Q is adjusted using resistor ratios in the summing amplifier. Equal resistor values are selected to maintain stable gain and Butterworth response characteristics.

The transfer functions are:

$$\text{LPF}(s) = \omega_0^2 / (s^2 + (\omega_0/Q)s + \omega_0^2)$$

$$\text{BPF}(s) = (\omega_0/Q)s / (s^2 + (\omega_0/Q)s + \omega_0^2)$$

$$\text{HPF}(s) = s^2 / (s^2 + (\omega_0/Q)s + \omega_0^2)$$

Component	Value	Purpose
R1	10 k Ω	Summing Amplifier
R2	33 k Ω	Feedback Network
R3	10 k Ω	Input Scaling
R4	10 k Ω	Gain Control
R5	10 k Ω	Gain Control
R6	16 k Ω	Integrator Stage
R7	16 k Ω	Integrator Stage
C1	10 nF	Frequency Control
C2	10 nF	Frequency Control

Table 7.1 – Component Values

8. SIMULATIONS

A) Proteus simulations were carried out to verify the filter behavior at different frequencies.

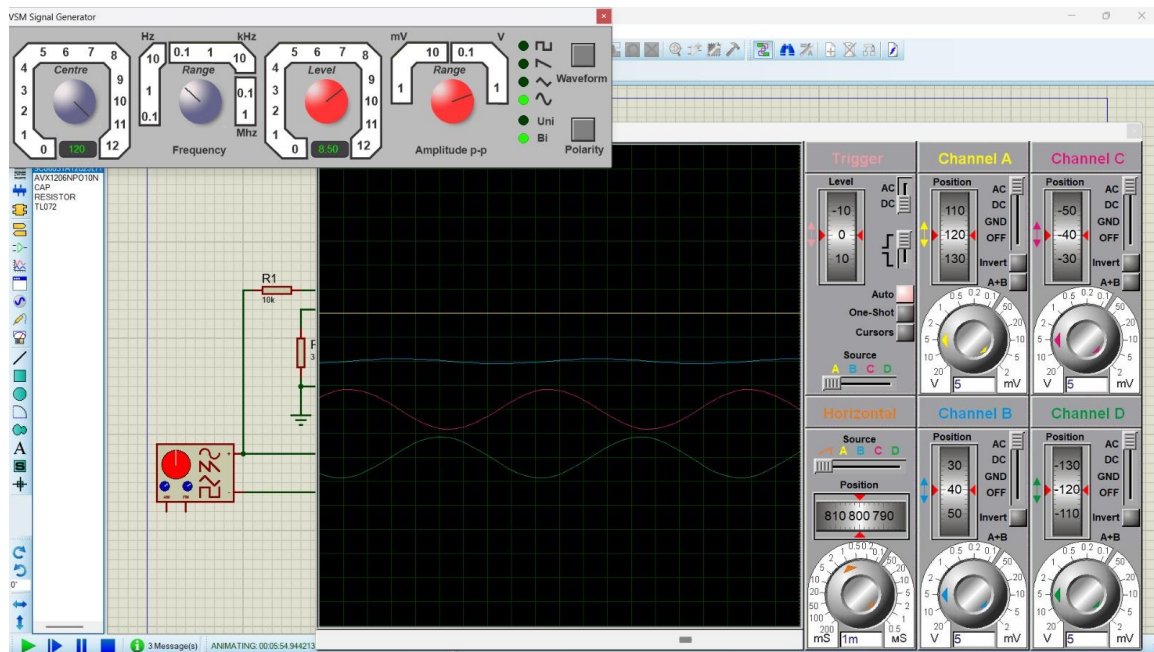


Fig 8.1 – Proteus Output at 120 Hz

Here, for frequency = 120Hz, only LPF(pink) output has a clear sine wave while HPF(yellow) output is a flat line and there is negligible output observed on BPF(blue). Note that the input is the green waveform

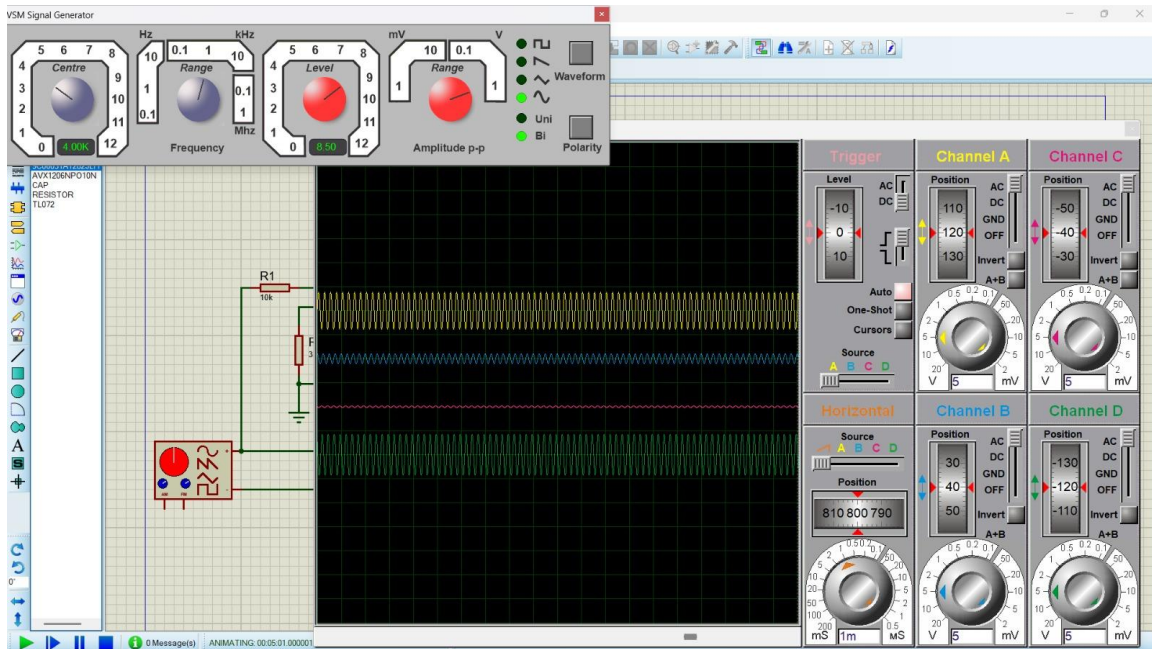


Fig 8.2 – Proteus Output at 1 kHz

Here, for frequency = 1KHz, all the three LPF,BPF AND HPF outputs are obtained as proper sine waves because at the central frequency all the three filters pass the input signal

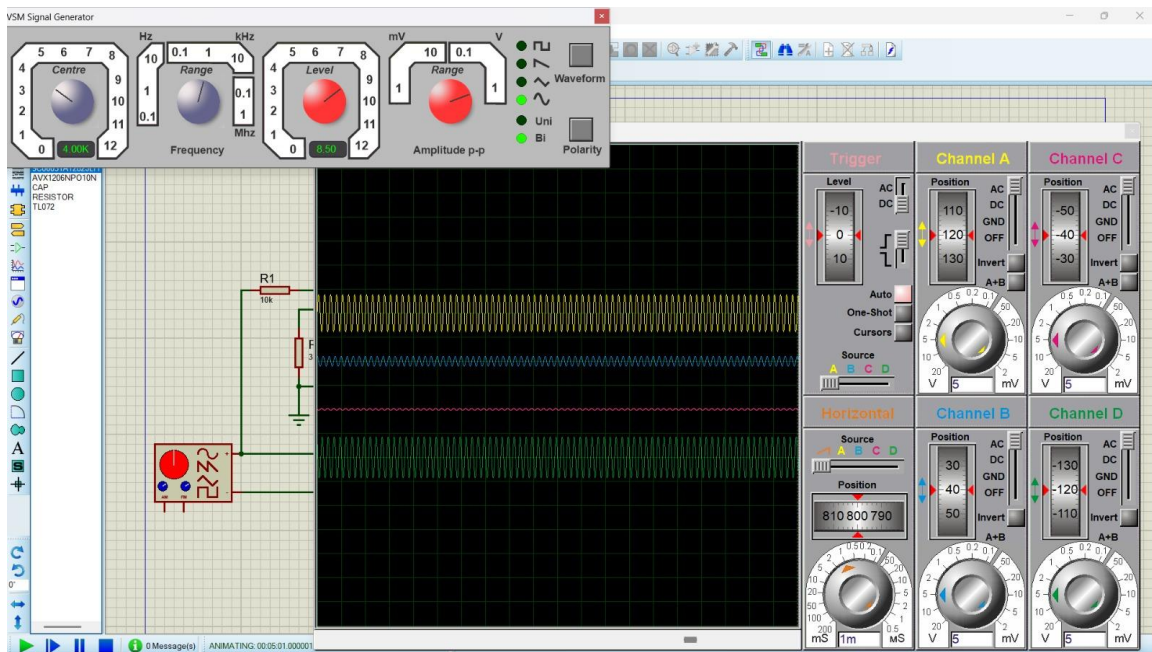


Fig 8.3 – Proteus Output at 4 kHz

Here, for frequency = 4KHz, only HPF provides a proper output signal since the BPF and LPF outputs for such a high frequency are almost fully suppressed, as seen in this simulation

B] Bode plots were generated in both dB and linear scales to verify the second-order frequency response.

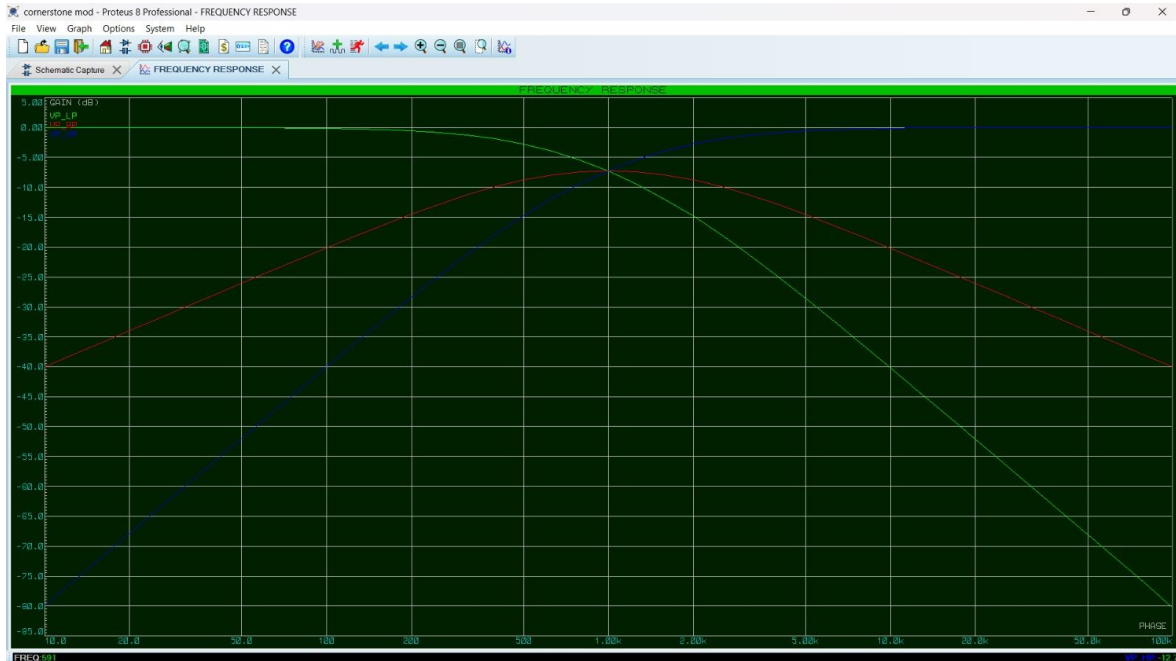


Fig 8.4 – Proteus Bode Plot (dB)

The lowpass output (green) is flat at 0dB from 10Hz up to about 500Hz, then hits -3dB exactly at 1kHz, and then rolls off steeply. Since this is a 2nd-order lowpass filter, the roll-off rate is -40dB per decade.

The highpass output (blue) mirrors this: it rises at +40dB/decade from low frequencies and flattens at 0dB above ~5kHz. Below 100Hz it reads around -75dB ; the two-pole high-pass roll-off rejecting low frequencies strongly.

The bandpass output (red) peaks around 1kHz (fc) at about -7dB, which tells you the peak gain is approximately $0.45\times$; slightly attenuated because it is only a single integrated BP node in the state-variable topology, not a dedicated resonator. It rolls off at -20dB/decade on both sides, giving it a broad, gentle passband.

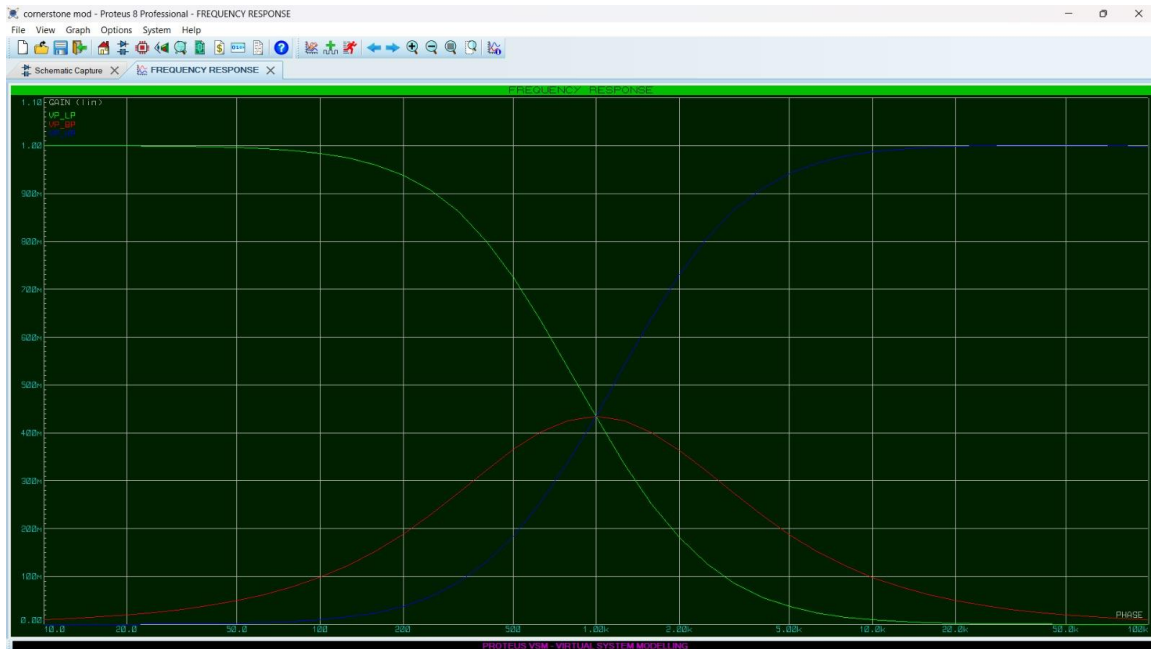


Fig 8.5 – Proteus Bode Plot (Linear)

Linear scale makes it even clearer visually. The lowpass (green) starts at gain = 1.0 (unity) and drops to 0.5 at 1kHz (which equals -3dB), then falls to nearly 0 by 10kHz.

The highpass (blue) does the exact opposite — starts near 0 at low frequencies and rises to unity above 5kHz.

The bandpass (red) peaks at about 0.42 gain at 1kHz and is symmetrically spread around that centre frequency. The point where LP and HP curves cross (both reading ~ 0.5 gain) falls exactly at 1kHz (cutoff frequency)

C) Step response analysis was also performed to study transient characteristics.

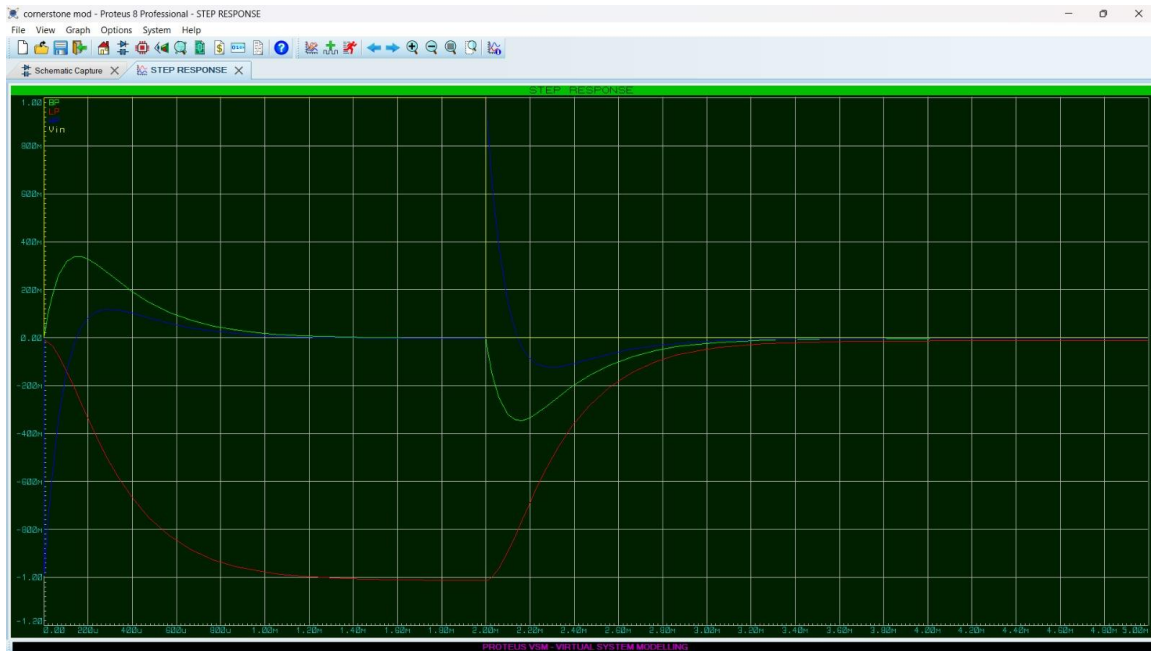


Fig 8.6 – Step Response of KHN Filter

Bandpass output (green): Before the step it shows a small positive hump peaking around +300mV, then decays back toward 0V. When the step hits at 2ms, it dips sharply to about -600mV and then slowly recovers back to 0V by 4ms.

Lowpass output (red): Before the step it slowly drifts near 0V, then at $t=2\text{ms}$ it follows the step downward, eventually settling toward -1V by 4–5ms.

*Highpass output (blue): Shows a sharp spike at the moment of each step edge, then immediately returns to 0V (the highpass only passes the instantaneous change, not the steady-state value).

9. HARDWARE IMPLEMENTATION AND OBSERVATIONS

The KHN filter circuit was implemented using TL071 operational amplifiers on hardware. Below is the image of the circuit implemented on breadboard

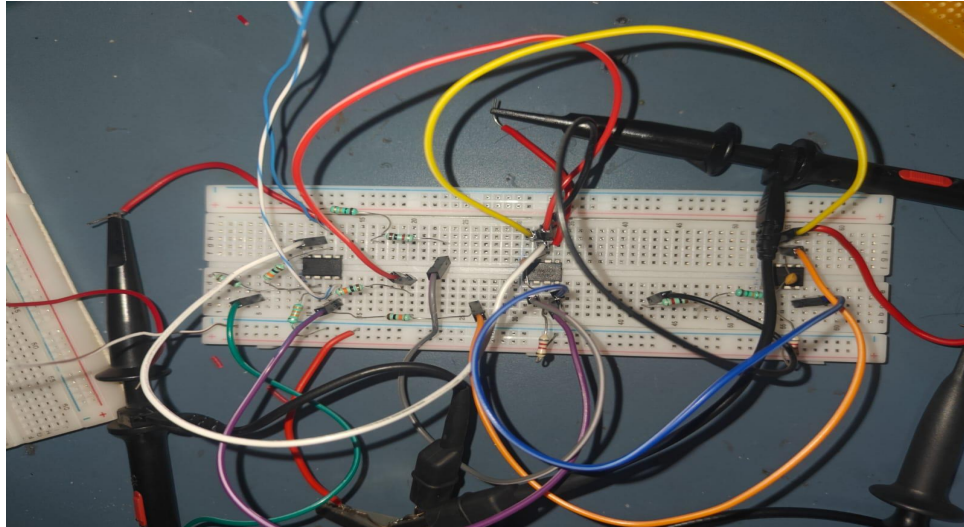


Fig 9.1 – Hardware implementation of circuit

We have further transferred this circuit onto zero PCB board by soldering for better representation, high reliability and strong wire connections

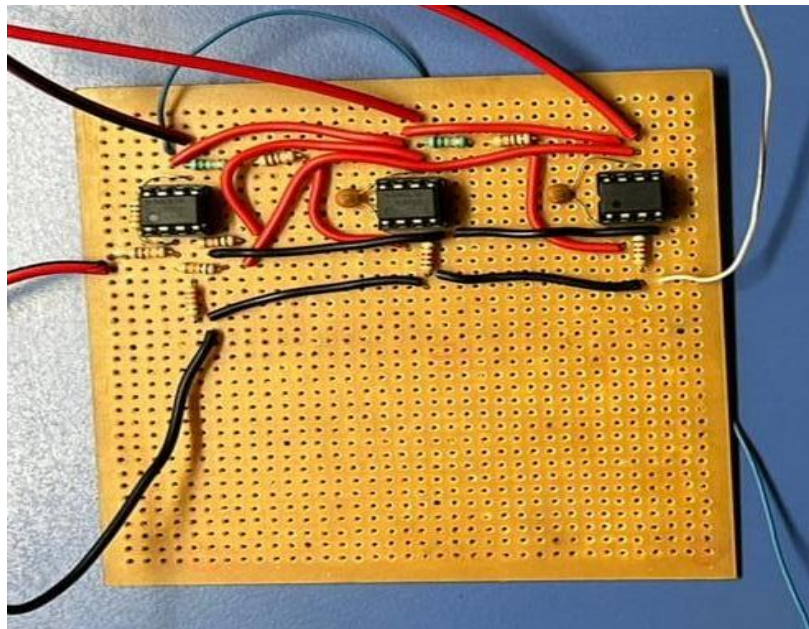


Fig 9.2 – Circuit implementation on zero PCB board

The outputs of the implemented circuit were observed using a Digital Storage Oscilloscope. Experimental results closely matched the Proteus simulation outputs. DSO measurements confirmed proper LPF, BPF, and HPF responses at different frequencies.



Fig 9.3 – DSO Output at 120 Hz

Here, we observe that at low frequencies the LPF output(dark blue) dominates, HPF output(light blue) is highly suppressed and the BPF output(pink) is also negligible. Note that the input signal is the yellow waveform

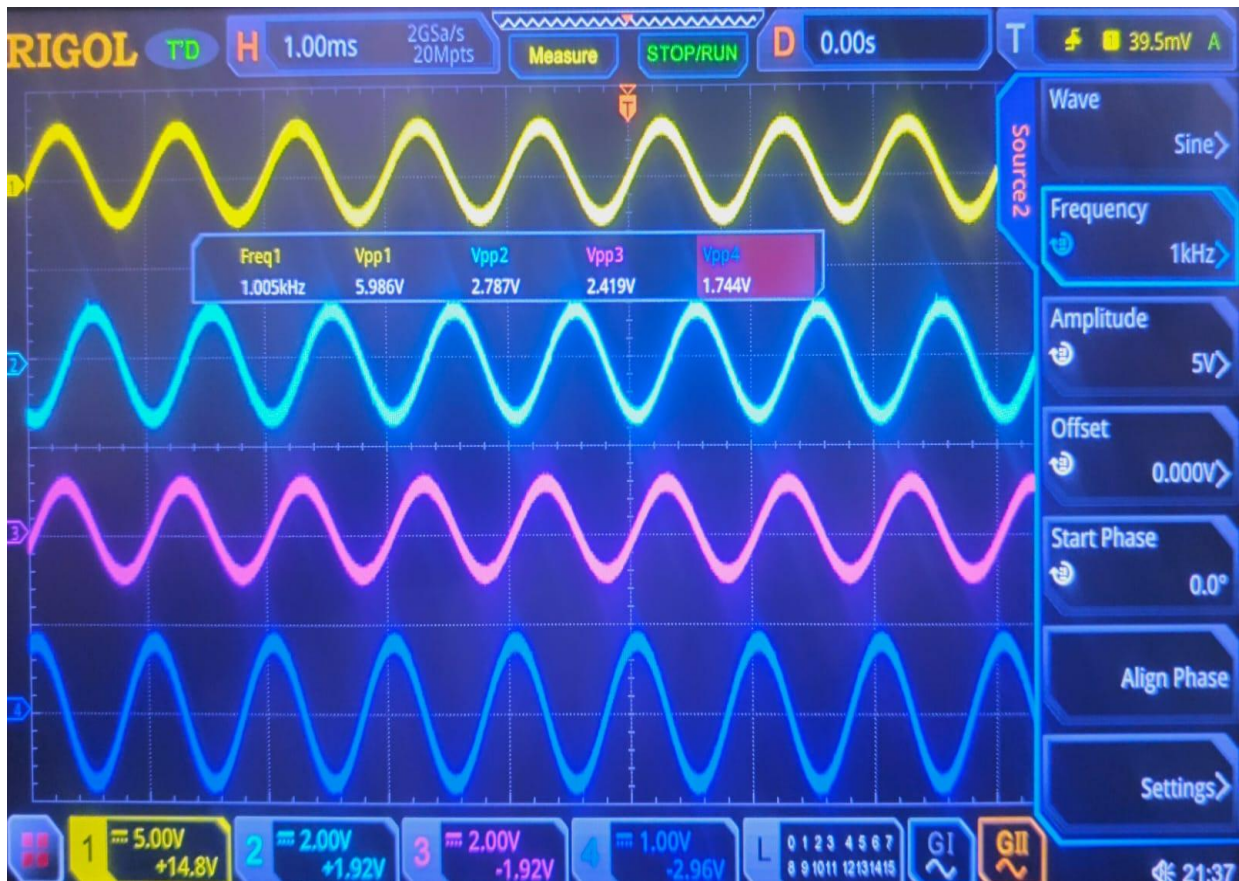


Fig 9.4 – DSO Output at 1 kHz

Here, the gain of all the filter outputs (LPF,HPF,BPF) are nearly equal, verifying that our cutoff frequency is close to 1KHz



Fig 9.5 – DSO Output at 4 kHz

Here, we observe that the HPF response dominates while BPF and LPF responses are comparatively fainter and weaker

Frequency ($V_{in} = 5.9\text{ V}$)	LPF Output	BPF Output	HPF Output
120 Hz	5.58 V	850 mV	126 mV
1KHz	2 V	2.5 V	2.75 V
4 KHz	300 mV	1.4 V	5.19 V

Table 9.1 – Summary of output voltages at different frequencies

10. COMPARISON OF SIMULATION AND HARDWARE RESULTS

The practical hardware implementation showed results very close to the Proteus simulations.

At 120 Hz, both hardware and simulation showed dominant low-pass response.

At 1 kHz, all three outputs were clearly visible and matched the expected behavior near the center frequency.

At 4 kHz, the high-pass response dominated in both simulation and practical testing.

Minor variations were observed due to resistor tolerances, op-amp non-idealities, and hardware noise.

Parameter	Simulation Result	Hardware Result	Observation
120 Hz Response	LPF output dominant	LPF output dominant	Simulation and hardware closely matched
1 kHz Response	All outputs visible near resonance	All outputs visible with slight amplitude variation	Expected center-frequency response achieved
4 kHz Response	HPF output dominant	HPF output dominant	Correct high-frequency response observed

Table 10.1 - Simulated vs Hardware results

11. APPLICATIONS

The KHN Audio Filter has several practical applications in analog and audio signal processing systems:

- Audio equalizers
- Music synthesizers
- Analog signal processing labs
- Biomedical signal conditioning
- Educational analog electronics experiments

12. CONCLUSION

The op-amp-based KHN filter provides a practical, reliable, and easily implementable second-order filter for audio applications. Unlike current-mode designs, this voltage-mode approach is ideal for laboratory prototyping, educational projects, and real-world analog circuits. The project demonstrated simultaneous LPF, BPF, and HPF outputs with a center frequency near 1 kHz. Both Proteus simulations and hardware testing confirmed the expected second-order response characteristics. The project highlights practical implementation of analog active filters using operational amplifiers.

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